

Chapter 3: Lipids

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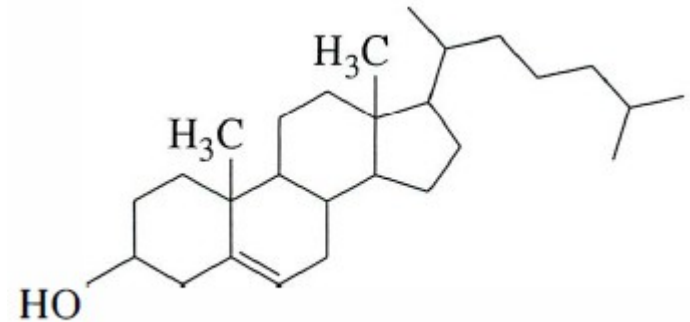
- 3.1. Introduction (Definition and classification)
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- 3.3. Triglycerides and waxes
- 3.4. Reactions of triglycerides (addition reactions, hydrolysis and oxidation [autooxidation, photooxidation and enzyme-induced oxidation])
- 3.5. Determination of analytical values for fats and oils (saponification value, acid value, iodine number, etc)

Lipids are broadly defined as

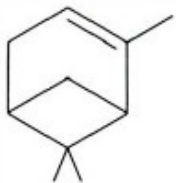
- ✓ **naturally occurring organic molecules that are soluble in non-polar solvents such as petroleum ether, benzene and chloroform.**
- ✓ **they are insoluble in water.**

– Includes

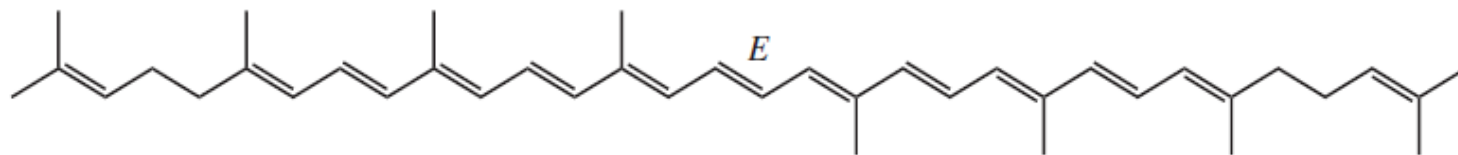
- Sterols
- Carotenoids
- Polyhydroxylalkanoates
- Triacylglycerols
- Terpenes



cholesterol, a steroid



α -pinene, a terpene

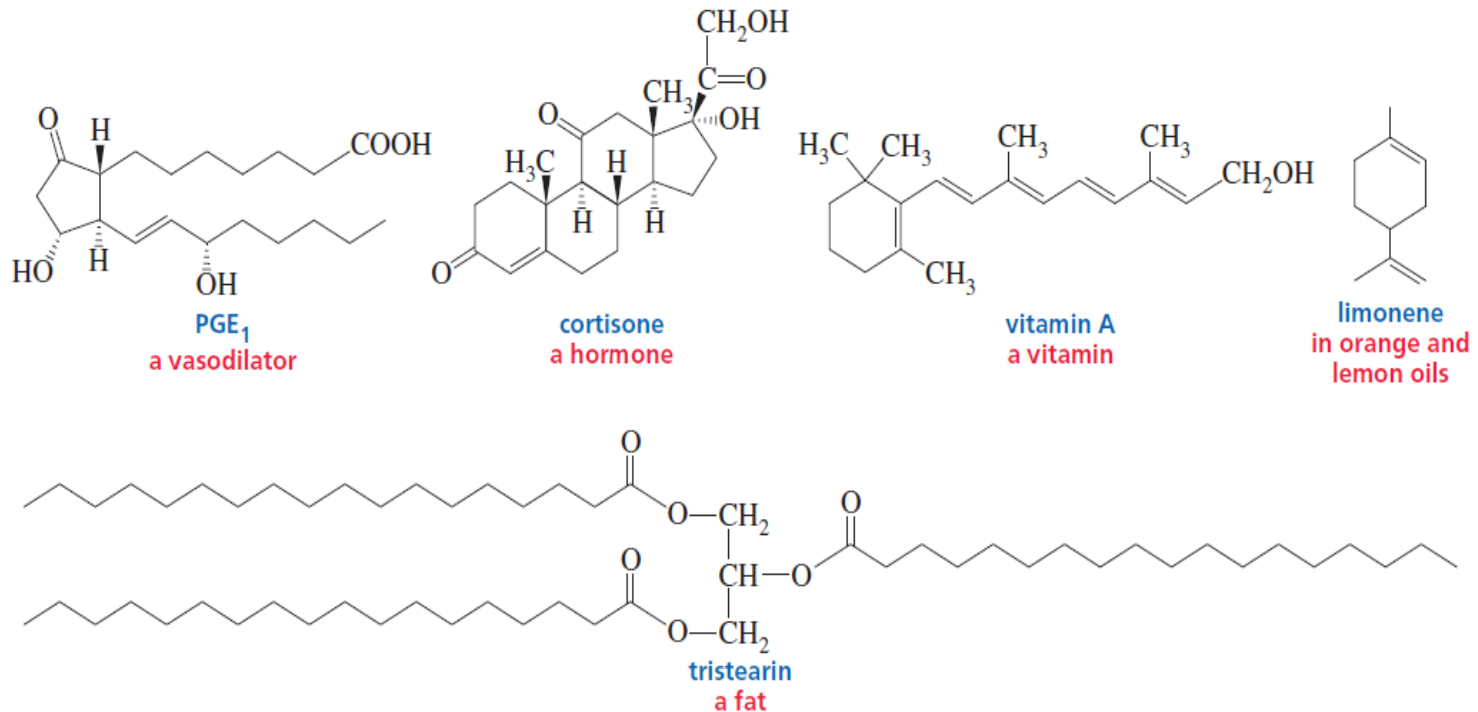


lycopene

- **Chemically**

- According to their ***physical properties rather than their structure.***

- Variety of structures and functions, as the following examples illustrate



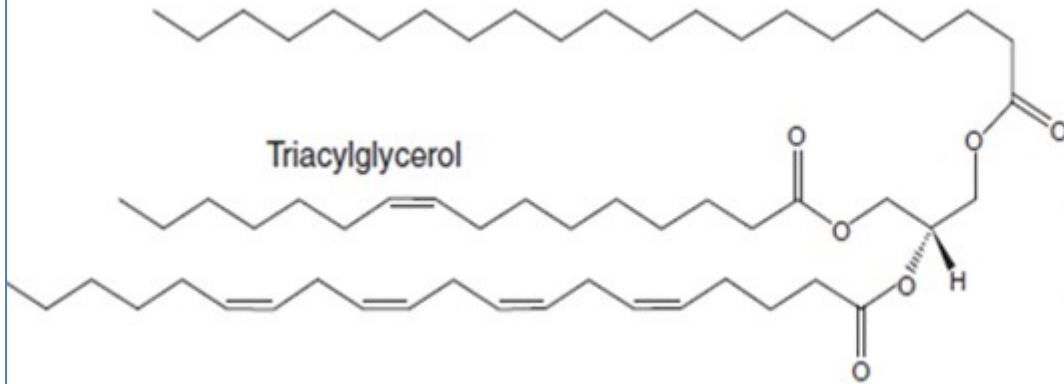
- **Classification:**
 - *Simple Vs Complex* lipids

1. Simple lipids

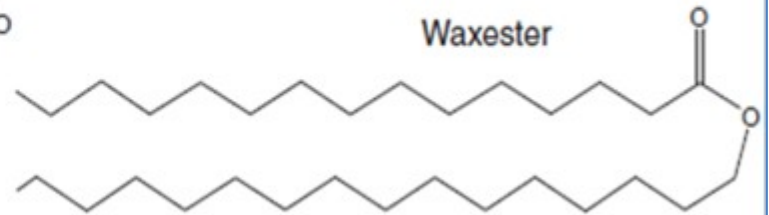
- Triglycerides or fats and oils are fatty acid esters of glycerol. Examples are lard, corn oil, cottonseed oil, and butter.
- Waxes are fatty acid esters of long-chain alcohols. Examples are beeswax, spermaceti, and carnauba wax.
- Steroids are lipids derived from partially or completely hydrogenated phenanthrene. Examples are cholesterol and ergosterol.

Simple lipids

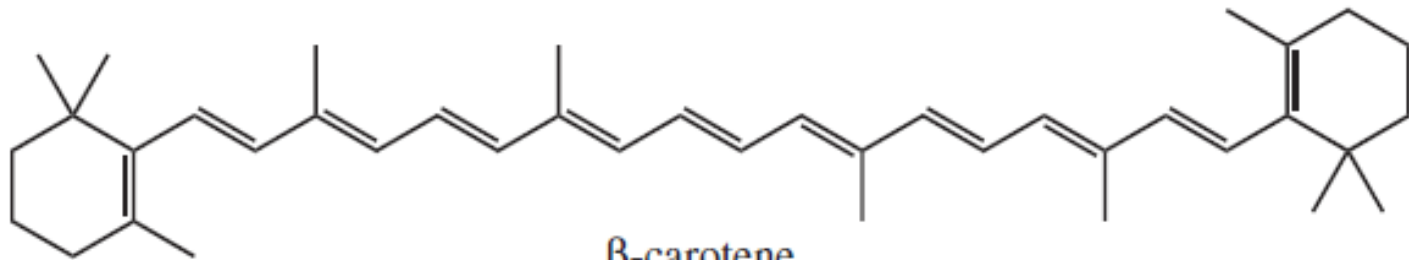
Triacylglycerol



Waxester



β -carotene



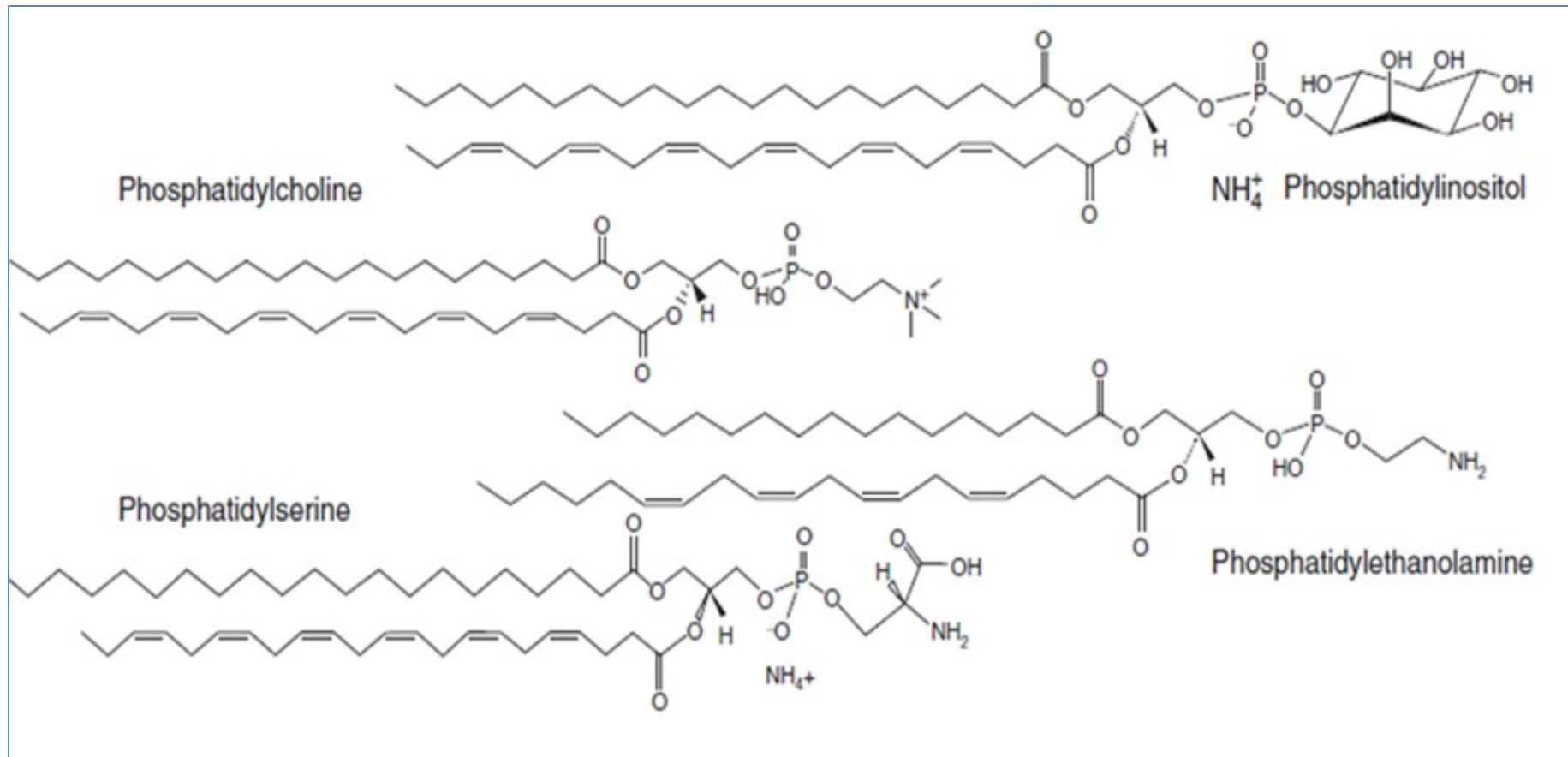
2. Complex lipids

- Phospholipids or Phosphatides are lipids which contain phosphorus and, in many instances, nitrogen. Examples are lecithin, cephalin, and phosphatidyl inositol.
- Glycolipids are lipids, which contain carbohydrate residues. Examples are sterol glycosides, cerebrosides, and plant phytoglycolipids.
- Sphingolipids are lipids containing the long-chain amino alcohol sphingosine and its derivatives. Examples are sphingomyelins, ceramides, and cerebrosides.

Complex lipids

Examples:

Phospholipids, glycolipids, ceramides, cerebroside and gangliosides



Functions

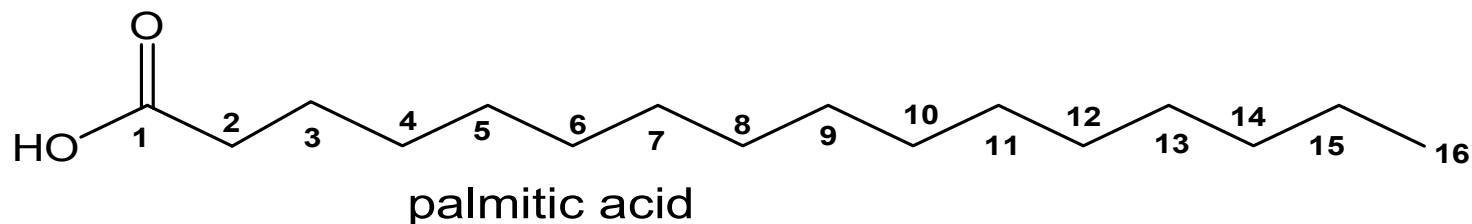
The lipids of physiological importance for humans have three major functions:

- 1.They provide energy reserves, predominantly in the form of triacylglycerols.
- 2.They serve as structural components of biological membranes.
- 3.Both lipids and lipid derivatives serve as vitamins and hormones.

3.2. *Fatty acids*

- **Fatty acids**

- Fatty acids are long-chain hydrocarbon molecules containing a carboxylic acid moiety at one end.
- Usually have an even number of carbon atoms, typically 12-24 C atoms.
- The numbering of carbons in fatty acids begins with the carbon of the carboxylate group.
- The melting point of fatty acids increases with increasing chain length



Cont...

- At physiological pH, the carboxylic acid group is readily ionized, rendering a negative charge onto fatty acids in bodily fluids.
- They are amphiphilic; they have a polar end and rest of the molecule is nonpolar.
- Fatty acids don't accumulate in cells. They are quickly converted into glycerides.
- More than 100 different fatty acids are known, and about 40 occur widely.

- **saturated or unsaturated**
 - Palmitic (C_{16}) and stearic acid (C_{18}) are the most abundant saturated fatty acids.
 - Oleic and linoleic acids (both C_{18}) are the most abundant unsaturated fatty acids.
- **according to chain length:**
 - short chain fatty acids : 2-4 carbon atoms
 - medium chain fatty acids : 6 –10 carbon atoms
 - long chain fatty acids : 12 – 26 carbon atoms
- **essential fatty acids vs those that can be biosynthesized in the body:**
 - linoleic and linolenic (infants develop lesions if non fat milk is given for long)

Nomenclature

1. Symbolic name: given as x:y ($\Delta^{a,b,c}$) where:
 - x is the number of C's in the chain,
 - y is the number of double bonds,
 - and a, b, and c are the positions of the start of the double bonds counting from C1 - the carboxyl C.
2. Systematic name using IUPAC nomenclature:
 - The systematic name gives the number of Cs
e.g. hexadecanoic acid for 16:0
 - If the fatty acid is unsaturated, the base name reflects the number of double bonds.
e.g. octadecenoic acid for 18:1 Δ^9
octadecatrienoic acid for 18:3 $\Delta^{9,12,15}$

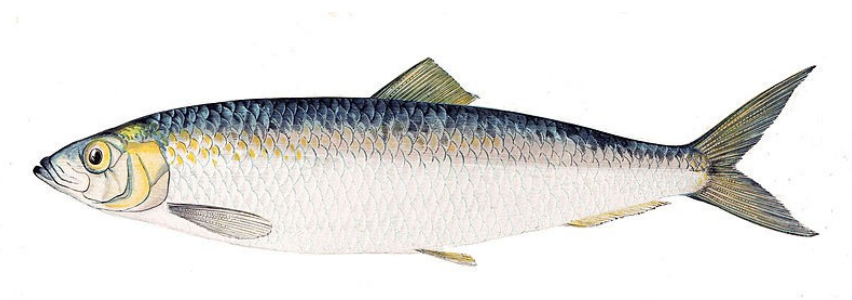
3. Common name:

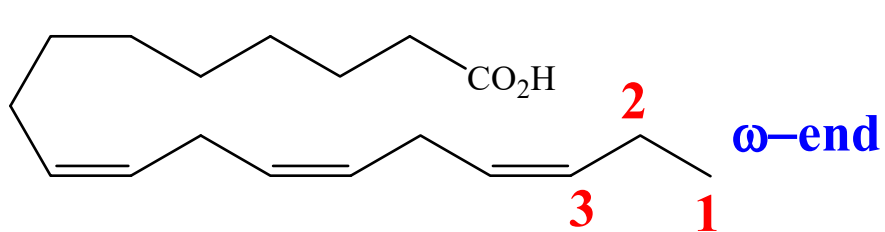
e.g. oleic acid, found in high concentration in olive oil

4. An alternative to the symbolic representation:

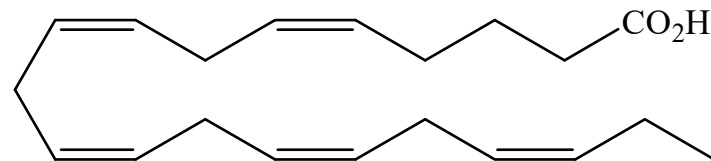
- The Cs are numbered from the distal end (the n or ω end) of the acyl chain (the opposite end from the carboxyl group).
 - Hence 18:3 $\Delta^{9,12,15}$ could be written as 18:3 (ω -3) or 18:3 (n-3) where the terminal C is numbered one and the first double bond starts at C3.
 - Arachidonic acid is an (ω -6) fatty acid while docosahexaenoic acid is an (ω -3) fatty acid.

- **ω -3 fatty acids (omega-3 fatty acids)**
 - α -linolenic acid, ***EPA***, DPA, and ***DHA***
 - ***Not found in seed oils of higher plants***
 - But in ***oily fish***, e.g. herring, tuna, mackerel
 - ***Reduce the risk of heart attacks and atherosclerosis***
 - ***DHA-Visual and neurological development in infants***

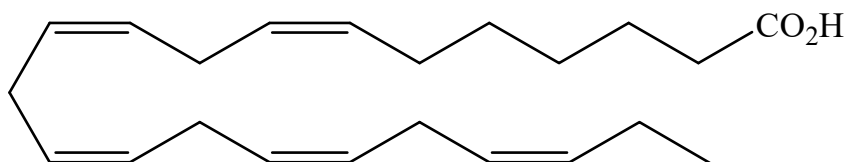




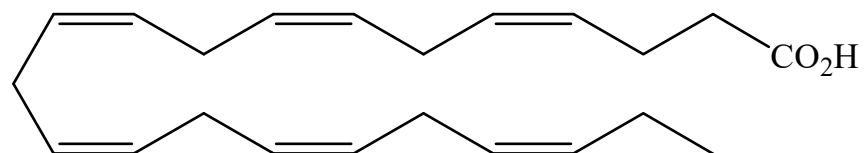
α -Linolenic
18:3 (9c,12c,15c)



Eicosapentaenoic (EPA)
20:5 (5c,8c,11c,14c,17c)



Docosapentaenoic (DPA)
22:5 (7c,10c,13c,16c,19c)



Docosahexaenoic (DHA)
22:6 (4c,7c,10c,13c,16c,19c)

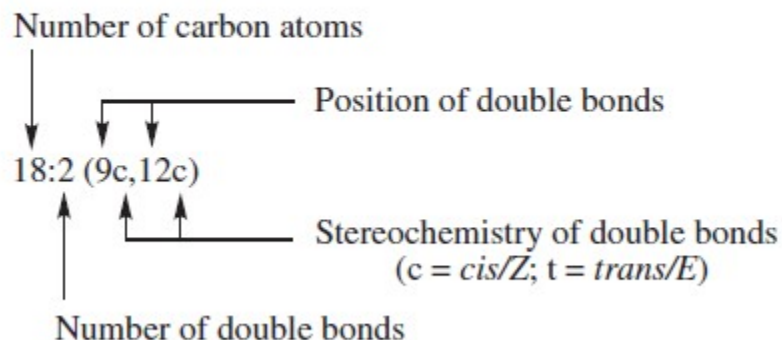
Common naturally occurring fatty acids

Saturated

butyric	4:0	
caproic *	6:0	
caprylic *	8:0	
capric *	10:0	
lauric	12:0	
myristic	14:0	
palmitic	16:0	
stearic	18:0	
arachidic	20:0	
behenic	22:0	
lignoceric	24:0	
cerotic	26:0	
montanic	28:0	
melissic	30:0	

* To avoid confusion, systematic nomenclature (hexanoic, octanoic, decanoic) is recommended

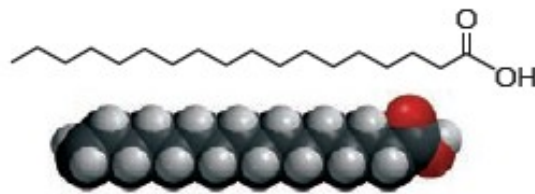
Abbreviations:



Unsaturated

palmitoleic		16:1 (9c)
oleic		18:1 (9c)
<i>cis</i> -vaccenic		18:1 (11c)
linoleic		18:2 (9c,12c)
α -linolenic		18:3 (9c,12c,15c)
γ -linolenic		18:3 (6c,9c,12c)
gadoleic		20:1 (9c)
gondoic		20:1 (11c)
arachidonic		20:4 (5c,8c,11c,14c)
eicosapentaenoic (EPA)		20:5 (5c,8c,11c,14c,17c)

- The *cis* configuration
 - ***Rigid bends or “kinks”***
 - Unsaturated fatty acids have ***lower melting points***



(a) Stearic acid

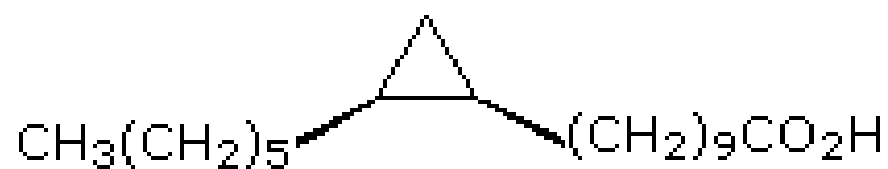


(b) Oleic acid

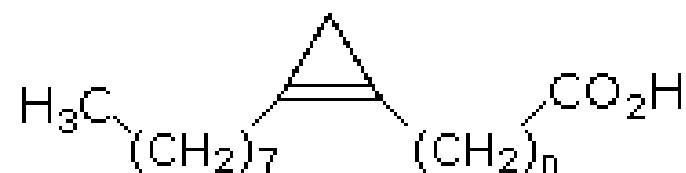
NAME	TYPICAL SOURCE	NUMBER OF CARBONS	NUMBER OF DOUBLE BONDS	CONDENSED FORMULA	MELTING POINT (°C)
Saturated					
Lauric	Coconut oil	12	0	$\text{CH}_3(\text{CH}_2)_{10}\text{COOH}$	44
Myristic	Butter fat	14	0	$\text{CH}_3(\text{CH}_2)_{12}\text{COOH}$	58
Palmitic	Most fats and oils	16	0	$\text{CH}_3(\text{CH}_2)_{14}\text{COOH}$	63
Stearic	Most fats and oils	18	0	$\text{CH}_3(\text{CH}_2)_{16}\text{COOH}$	70
Unsaturated					
Oleic	Olive oil	18	1	$\text{CH}_3(\text{CH}_2)_7\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}(\text{cis})$	4
Linoleic	Vegetable oils	18	2	$\text{CH}_3(\text{CH}_2)_4\text{CH}=\text{CHCH}_2\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}(\text{all cis})$	-5
Linolenic	Soybean and canola oils	18	3	$\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{CH}=\text{CHCH}_2\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}(\text{all cis})$	-11
Arachidonic	Lard	20	4	$\text{CH}_3(\text{CH}_2)_4(\text{CH}=\text{CHCH}_2)_4\text{CH}_2\text{CH}_2\text{COOH}(\text{all cis})$	-50

Less Common Fatty Acids

- A wide array of unusual fatty acids is found in seed oils and microorganisms such as bacteria.
- Among these are fatty acids are:
 - cis-cyclopropane fatty acids (e.g. lactobacillic acid),
 - omega-cycloalkyl fatty acids (e.g. 2-hydroxy-11-cycloheptylundecanoic acid)
 - cyclopropene fatty acids (e.g. malvalic and sterculic acid).

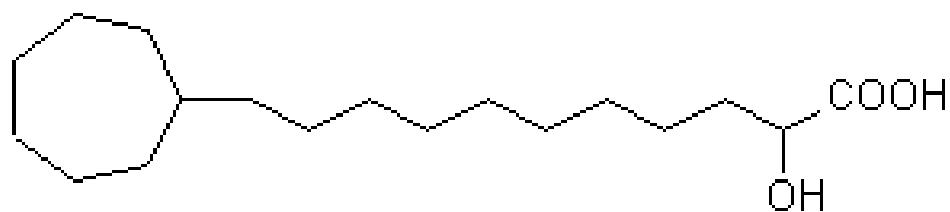


lactobacillic acid



$n = 6$ Malvalic Acid

$n = 7$ Sterculic Acid

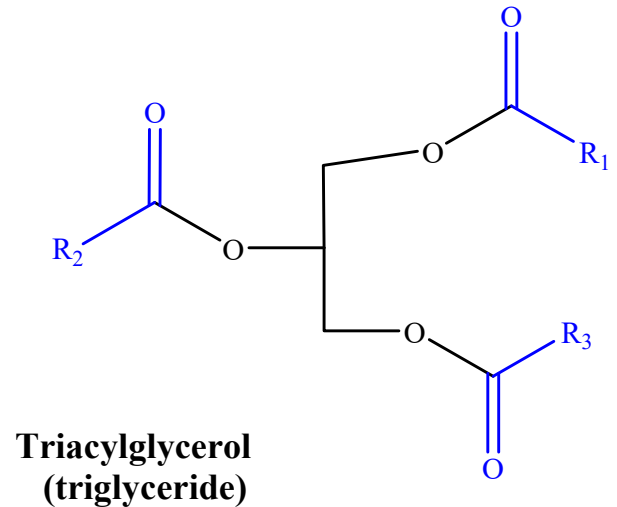


2-hydroxy-11-cycloheptylundecanoic acid

3.3. Triglycerides (fats and oils) and waxes

- **Triglycerides (triacylglycerols)**

- Esters of fatty acids and glycerol
- Solid or liquid at room temperature:
 - Fixed/edible oils
 - Fats
- Animal vs plant TAGs



Fats

- Fats

- mostly from animal sources,
- have all single bonds between the carbons in their fatty acid tails,
- thus all the carbons are also bonded to the maximum number of hydrogens possible.

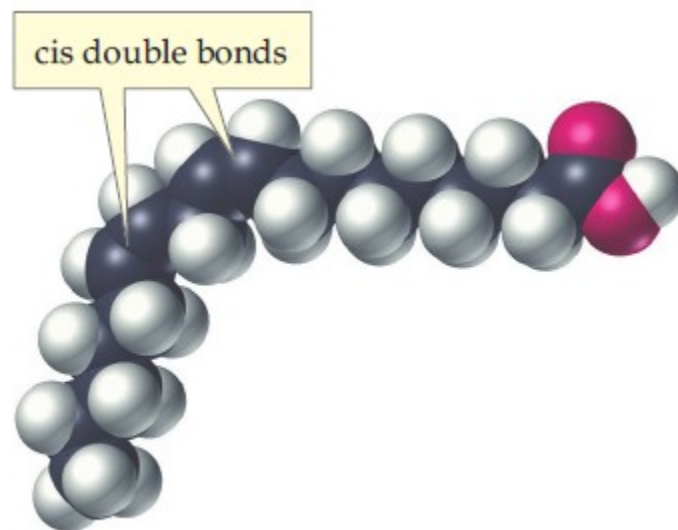
- Since the fatty acids in these triglycerides contain the maximum possible amount of hydrogens, these would be called **saturated fats**.
- The hydrocarbon chains in these fatty acids are, thus, fairly straight and can pack closely together, making these **fats solid at room temperature**.

Oils

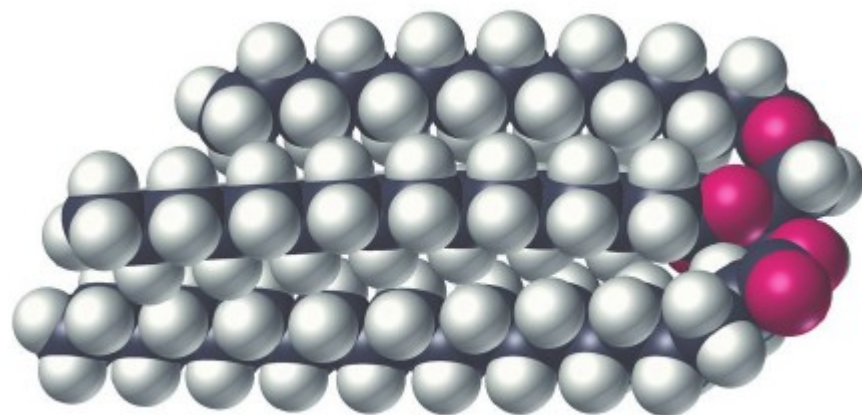
- Oils,
 - ✓ mostly from plant sources,
 - ✓ have some double bonds between some of the carbons in the hydrocarbon tail, causing bends or “kinks” in the shape of the molecules.
- Because some of the carbons share double bonds, they’re not bonded to as many hydrogens as they could if they weren’t double bonded to each other. Therefore these oils are called **unsaturated fats**.
- Because of the kinks in the hydrocarbon tails, unsaturated fats (or oils) can’t pack as closely together, making them *liquid at room temperature*.



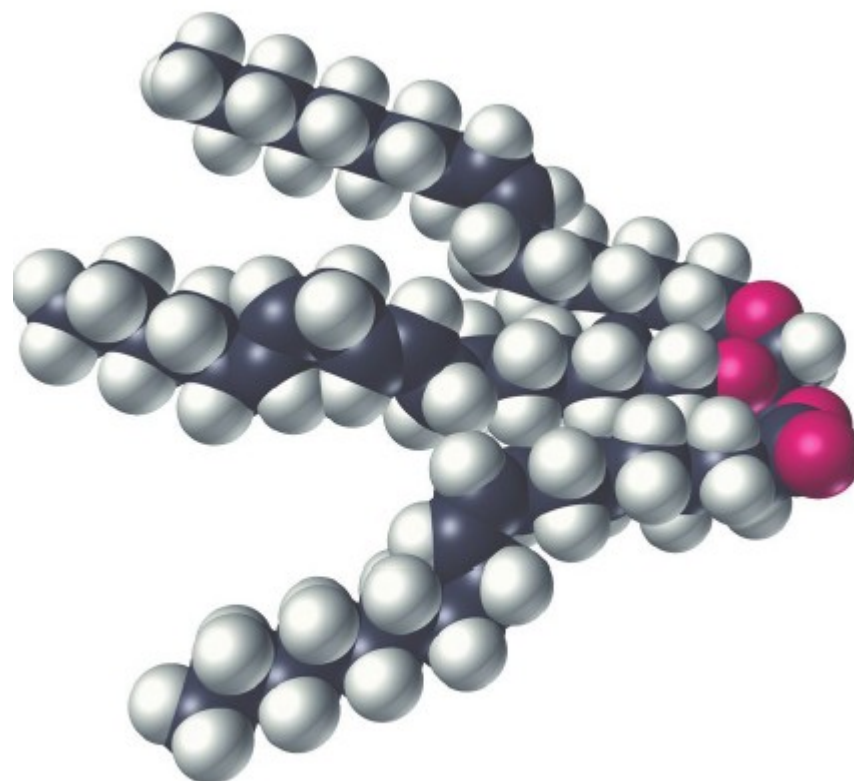
Stearic acid, an 18-carbon saturated fatty acid



Linoleic acid, an 18-carbon unsaturated fatty acid

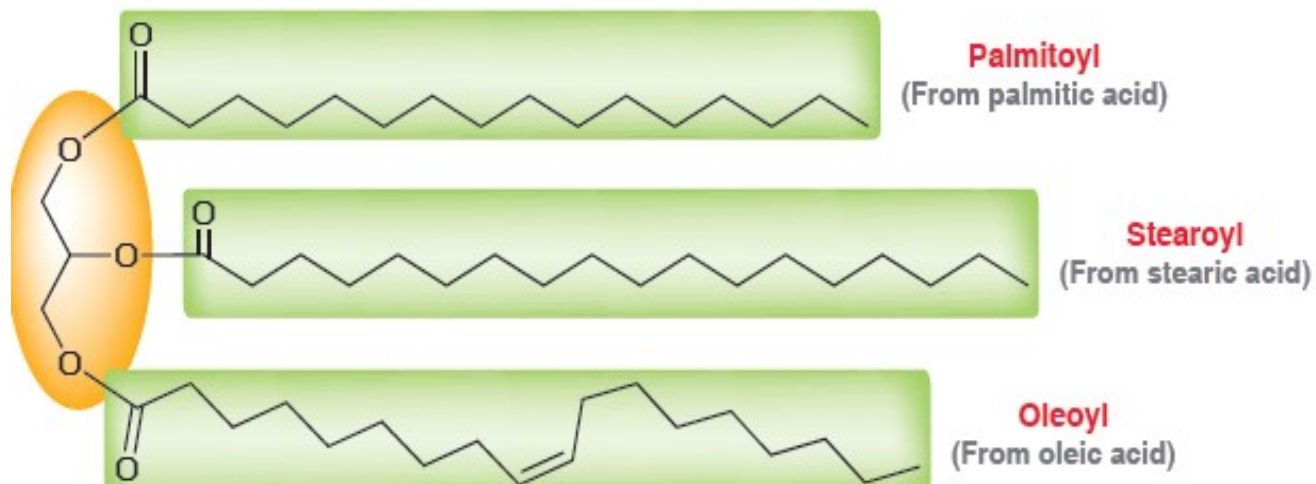


A fat



An oil

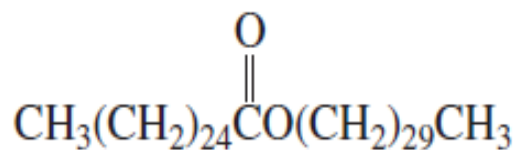
- Based on the ***type*** of esterifying FAs:
 - **Simple triglyceride:**
 - it gives only one type of fatty acid on hydrolysis
 - **Mixed triglyceride:**
 - It gives different types of fatty acid on hydrolysis



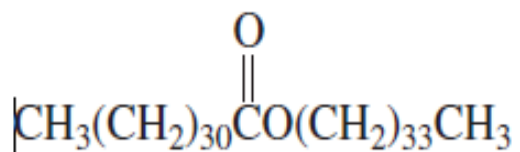
- **Waxes**

- ***Esters of long-chain carboxylic acids and long-chain alcohols***

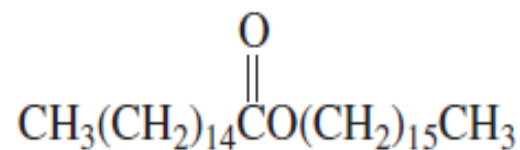
- The carboxylic acid: 16 – 36 carbons
 - The alcohol: 24 - 36 carbons



a major component of
beeswax
structural material
of beehives



a major component of
carnauba wax
coating on the leaves
of a Brazilian palm

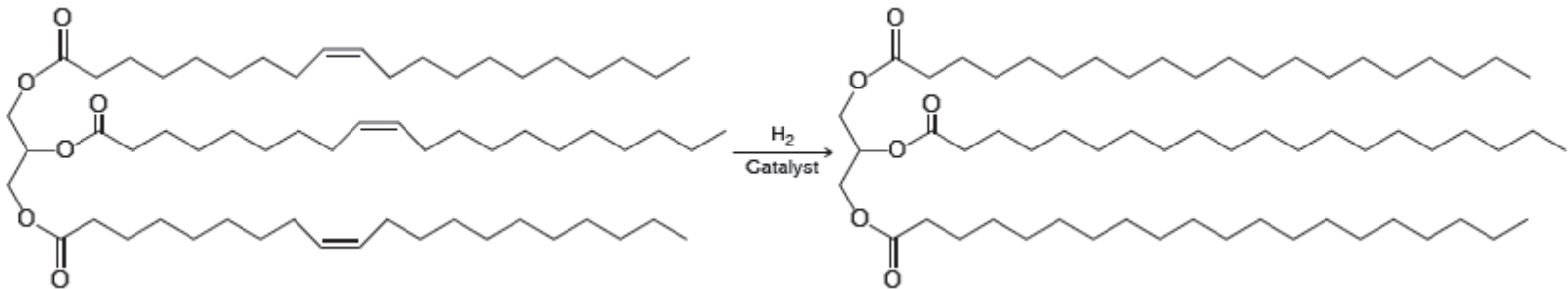


a major component of
spermaceti wax
from the heads of
sperm whales

3.5. Reactions of fats and oils

1. Hydrogenation ($\text{H}_2/\text{Pd}/\text{C}$)

- Converts unsaturated fatty acids to saturated FA
 - $\uparrow$ Viscosity
 - E.g. Margarine = hydrogenation product of soybean, peanut, or cottonseed oil



2. Dehydrogenation (chromium (III) oxide)

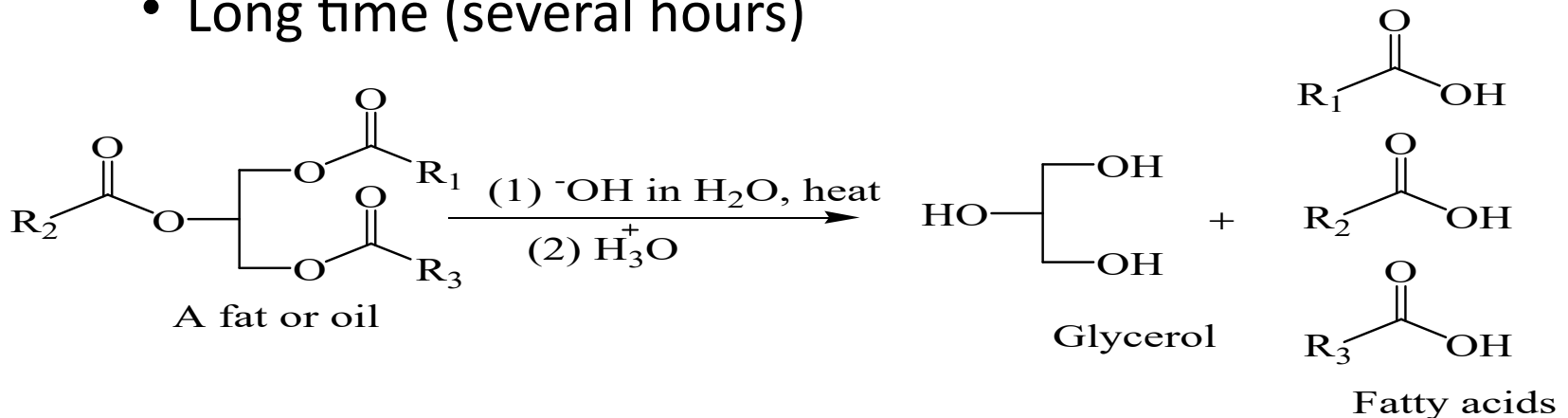
- ✓ converts saturated fatty acid to unsaturated FA

3. Halogenation

Chlorine, bromine, iodine

4. Hydrolysis

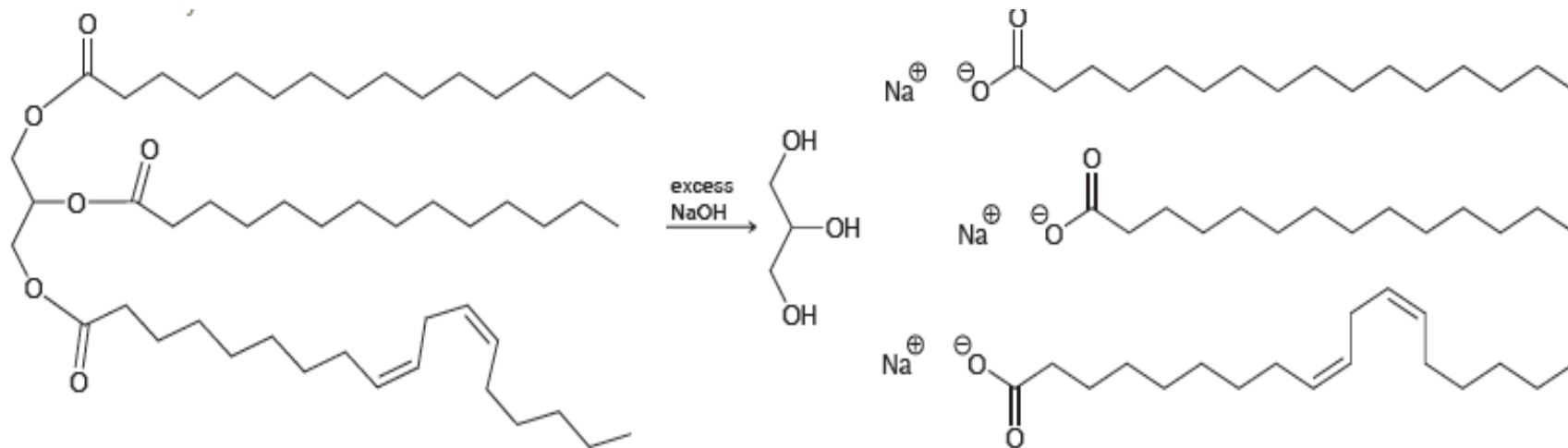
- Reaction of *fats and oils with water in the presence of either acid or base*
- Requires
 - Fat-soluble catalyst
 - High temperature ($>100^{\circ}\text{C}$)
 - Long time (several hours)



4. Saponification (Soap formation)

– *Sodium or potassium salts of fatty acids*

- *Saponification*



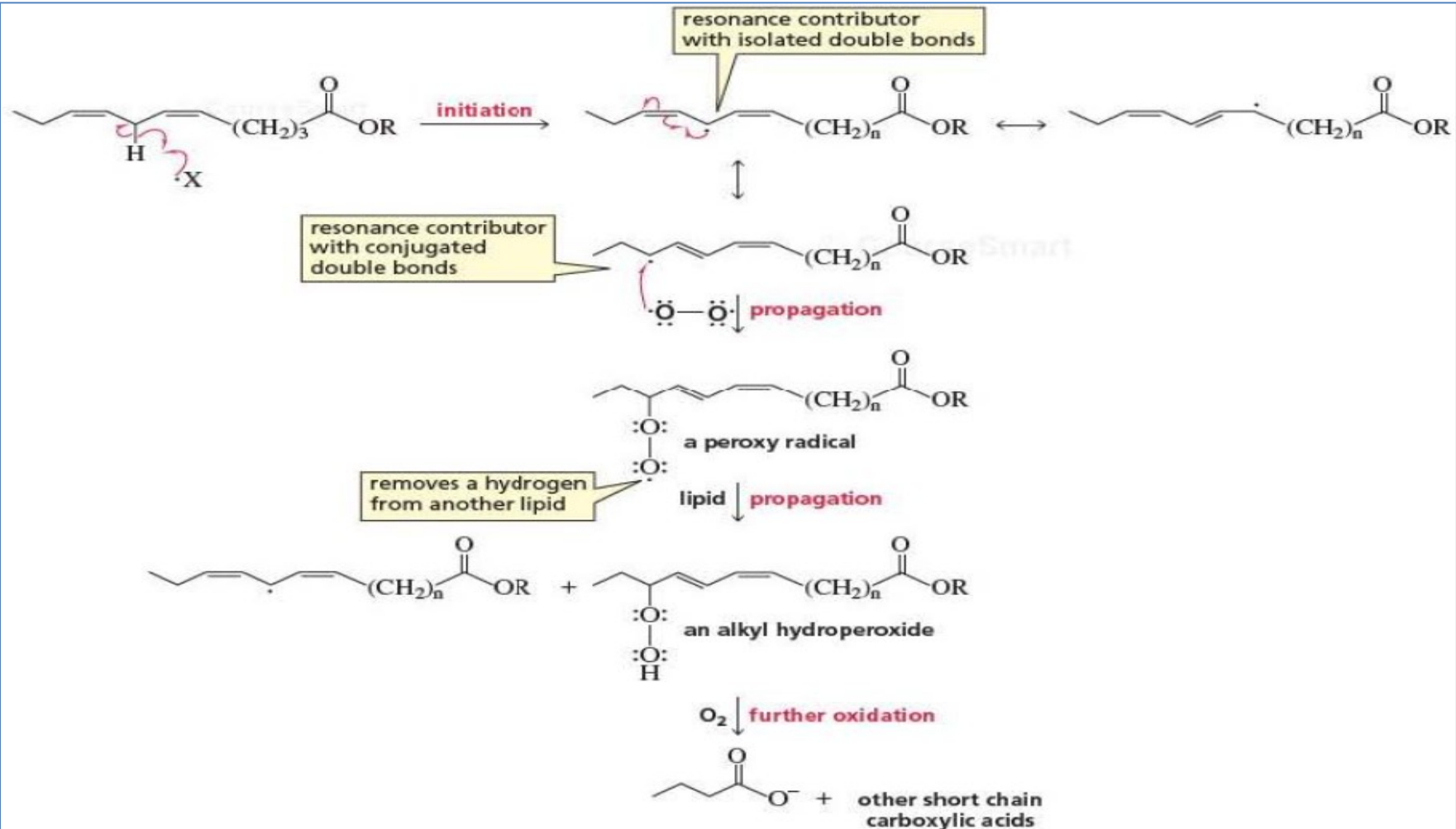
5. Oxidation

- **Oxidative rancidity**: the *deterioration* of a fat or oil in the *presence of oxygen*
- Initially, addition of oxygen at or near the double bond of a fatty acid chain

- Three types of lipid oxidation:
 - **Autoxidation,**
 - **Photo-oxidation (or photooxygenation), and**
 - **Enzyme-catalyzed oxidation**

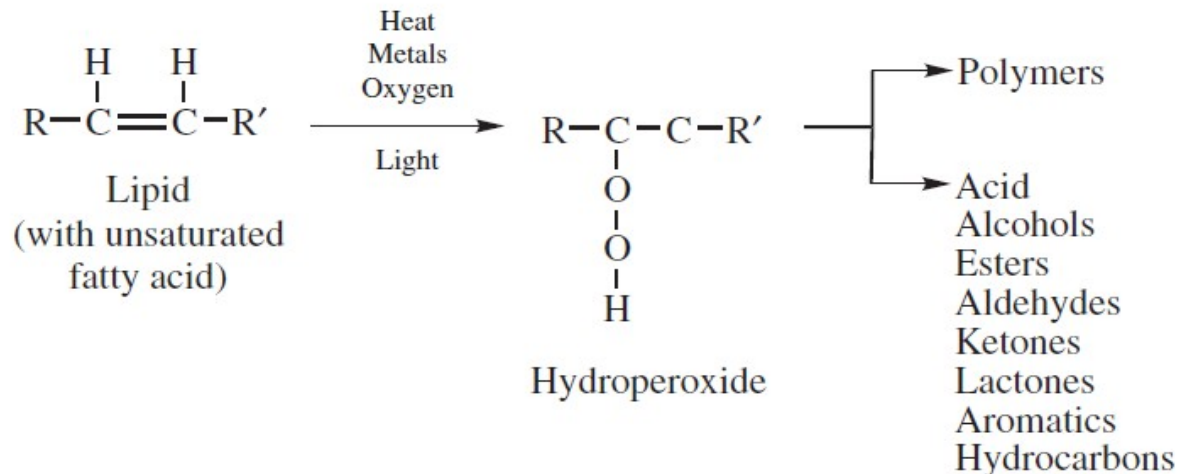
A. Lipid autooxidation

- ✓ Free radical chain reaction
- ✓ Free radicals



B. Photo-oxidation

- Addition of *singlet oxygen* to the double bond by the *concerted “ene” addition mechanism*, and
- Hydrogen atom transfer from a *neighbouring –CH₂ – group* to the terminal oxygen leads to the formation of *hydroperoxy* lipids
- Not a chain reaction



C. Enzyme-catalyzed oxidation involves

- ***Enzymatic production*** of peroxides
- ***Lipases and lipoxygenases***

- **Oxidative rancidity**

- Effect on:

- **Color**

- Colored compounds of *quinoid nature*

- » E.g. Cottonseed and soybean oil are subject to marked darkening upon oxidation

- **Odor and taste**

- Butter -Butyric acid

- Sour milk-lactic acid

3.6. Determination of analytical values for fats and oils

3.6.1. *Saponification value*

- **Principle**

- Saponification: Hydrolysis of fat with an alkali
 - Salts of fatty acids (also called soap) and glycerol.
- Saponification value:

“mg of KOH required to saponify 1g of the given fat”

- *Calculation:*

- 0.5 N KOH in blank = ***x*** mL
- 0.5 N KOH in test sample = ***y*** mL
- Titre value for sample = (***x—y***) mL

$$\text{Saponification value} = \frac{28.5 \times \text{titre value}}{\text{Weight of sample (g)}}$$

3.6.2. *Acid value*

- **Fats and oils**

- Become *rancid during storage*
- The amount of free fatty acids can be determined *volumetrically by titrating the sample with potassium hydroxide*
- Age and extent of its deterioration
 - Acid value

- Acid value:
“mg of KOH required for neutralizing the free fatty acids present in 1g of fat and oil”
- *Calculations*
 - 0.1 N KOH solution used for blank = ***x*** mL
 - 0.1 N KOH solution used for sample = ***y*** mL
 - Titre value for sample = ***(y—x)*** mL

$$\text{Acid value} = \frac{\text{Titre value} \times \text{Normality of KOH} \times 56.1}{\text{Weight of sample (g)}}$$

3.6.3. Iodine number

- *Calculations*

- Volume of 0.1 N sodium thiosulfate used for blank = ***x*** mL
- Volume of 0.1 N sodium thiosulfate used for sample = ***y*** mL

$$\text{Iodine number} = \frac{(x - y) \times 12.7}{1000} \times \frac{100}{\text{Weight of sample (g)}}$$

- **How do we know degree of carbon to carbon unsaturation of fat, oil or their derivatives?**
 - Iodine value or iodine number:
 - “ ***grams of iodine absorbed by 100 g of fat***”
 - Degree of oxidative rancidity